

Integration Notes - 2026-09-19

Source basis: Perplexity scouting cycle **AI-SLOP-research-cycle-2026-09-19**, compared against **merrypranxter/ai_slop** at commit 63e887551e6e19dab35e3b287a2226e91f1d0e17. These notes are a project-routing summary, not a substitute for the full scouting report or its ledger.

11 selected findings	0 new canonical operators	2 major synthesis chains	1 first build target
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What changed

The useful result is not another pile of new names. The cycle found concrete implementations, validators, regulators, and adapters that make existing AI SLOP ideas executable, measurable, or harder to fool. The project should resist ontology inflation: **Protocol Speciation** and **Protocol-Mismatch Collective** remain candidate procedural designs; **Silent Scar** is an experiment label/validator, not a new memory operator.

The chain to build first

1. PERCEPTUAL WOUND	Lenia sensory occlusion changes what an organism can sense without deleting matter.
2. SILENT SCAR	After visible recovery, probe the system again to test whether hidden dynamics or future response changed.
3. BEHAVIOR ARCHIVE	Use AutoQD-style descriptors based on intervention-response traces, not just visual similarity.

Why this matters: it turns the project's "artifact remembers the wound" idea into a controlled experimental pipeline. The visible form can look recovered while the future behavior still carries history. That is exactly the kind of scar AI SLOP has been talking about, now with a test.

Second strong chain

Protocol evolution -> codebook mismatch -> transmission test. GlossoGen supplies pressure-driven compact communication plus newcomer swaps; the communication-heterogeneous NCA work supplies mismatched internal codebooks under a shared objective. Together they suggest a system where a private protocol must first become useful, then survive translation damage, then be learned by a replacement partner. This is much stronger than asking a model to invent alien gibberish.

The 11 selected findings

Finding	Role in AI SLOP	E / U	Status / immediate note
Lenia sensory occlusion	Perception-only wound; separate sensed world from physical state	E4 / U5	BUILD FIRST; public simulation repo
Glossogen	Pressure-driven private protocol + newcomer transmission test	E3 / U5	Candidate Protocol Speciation; MIT platform
MusicLayout	Editable intermediate score / temporal law before rendering	E3 / U5	Adapter; code public, trained checkpoint unclear
Latent-Control Heads (LatCH)	Drive audio with measurable latent target curves	E3 / U5	Adapter; demo/paper confirmed, release incomplete
Diff2Mix	Explicit per-track production/effects parameters	E3 / U4	Adapter; demos + differentiable console, full checkpoint unclear
NCA attractor analysis	Hidden-dynamics recovery test / Silent Scar validator	E4 / U5	BUILD EARLY; public notebooks/models
Communication-heterogeneous NCA	Conflict from incompatible codebooks despite shared goal	E3 / U5	Candidate Protocol-Mismatch Collective; implementation not yet found
AutoQD	Behavioral descriptors from interaction traces for quality-diversity archives	E4 / U5*	BUILD WITH LENIA; *transfer-to-art evidence still E2
SAE random-baseline sanity checks	Research hygiene: learned control vs random-direction competitors	E4 / U5	Validator, not generator
Stable Video Infinity	Generator-specific error bank; repair some failures, preserve chosen scars	E4 / U4	Regulator; public ecosystem/weights
IDAttn	Timed isolation/reunion of attention jurisdictions	E3 / U4	Implementation lead; code/checkpoint not established

Important: what NOT to promote

Do not turn any of the following into project claims: "infinite video never drifts," "visual recovery proves full recovery," "a named SAE feature proves a native concept," or "emergent gibberish proves a language." The report explicitly rejects those upgrades. Creative utility and mechanism confidence stay separate.

Repository integration notes

Route	What to do
ADD	Perception-only perturbations; full-state recovery probes; behavior-sensitive archive descriptors; explicit music-layout / latent-control / effects-parameter adapters; model-specific error-bank regulator; timed isolation/reunion implementation lead.
UPDATE	Topic 01 mechanistic-interpretability controls with learned-vs-random SAE comparisons; Topic 02 joint-attention routing and edit-occurrence vs edit-correctness distinction; attractor/memory docs with full-state dynamics and second-probe tests; video docs with identity/motion tradeoffs.
MERGE	Alien failure-profile material into Alien Distance Metric examples. Protocol Speciation stays under Symbolic Compression + Token Tax + Separated Jurisdictions. Protocol-Mismatch Collective stays under Separated Jurisdictions + Forced Aliasing. Silent Scar stays under Hysteretic Transformation / Artifact Fossilization as a validation protocol.
ARCHIVE / REJECT	Claims and labels whose only novelty is metaphor, repo naming, or a port. Anthropomorphic interpretations of simulated avoidance. "Winning prompt language" claims without controlled evidence.
INVESTIGATE	MusicLayout trained weights; LatCH released heads; Diff2Mix full release; IDAttn implementation; SAE evaluation harness; prior canonical scouting ledger / missing scouting skill provenance.

Proposed build order from this cycle

- 1. Lenia occlusion prototype.** Reproduce no-mask vs state-erasure vs unnormalized mask vs normalized sensory occlusion. Log heading, position, connected components, overlap, and recovery.
- 2. Silent Scar assay.** Select visually recovered runs, apply the same second injury, and compare the full trajectory to matched uninjured controls.
- 3. AutoQD behavior archive.** Compare image embeddings, hand-designed recovery descriptors, and learned trace descriptors. Preserve history/scar state when it changes future behavior.
- 4. Protocol evolution experiment.** Small/large communication budget x postmortem/no-postmortem; test unseen combinations and a replacement partner; include a hand-designed abbreviation-code control.
- 5. Protocol mismatch experiment.** Use reversible codebooks/rotations and separately ablate vote inversion. Measure message amplitude, consensus time, and spatial fragmentation.
- 6. Media adapters after the core works.** MusicLayout for editable temporal laws; LatCH for measurable target curves; Diff2Mix for explicit production mutations.

The project's guardrail for future scouting

A finding graduates in stages: **candidate -> implementation -> supported project mechanism -> new operator**. A runnable intervention is required before "implementation"; a matched comparison is required before "supported"; and a new operator only exists if the capability cannot already be represented as an implementation, composition, regulator, or validator of something we have. Preserve negative results and reasons when revising or archiving.

Compact working memory

KEEP IN HEAD

Perceptual wounds can leave behavioral scars.

Do not equate visible recovery with dynamical recovery. Probe again.

Archive behavior, not just appearance.

A weird artifact is more interesting when its response to intervention is distinct and reproducible.

Private languages must earn their weirdness.

Coordination success + generalization + newcomer transmission beat gibberish aesthetics.

Interpretability knobs need stupid baselines.

If a random direction works just as well, keep the creative knob if useful - demote the story about what it 'means.'

Repair and scar preservation are separate objectives.

Error banks can teach a generator what to recover from while the project explicitly exempts chosen artifacts from repair.

Separated jurisdictions need permeability schedules.

Permanent isolation can destroy coherence; controlled reunion timing is itself an experimental variable.

Release blockers / unresolved questions

MusicLayout trained weights; LatCH control-head release; full Diff2Mix checkpoint; IDAttn code/checkpoint; SAE evaluation harness; causal confounds in protocol mismatch; spatial blindness in sorted-profile recovery metrics; shuffled-layout gains; motion-vs-freezing in video consistency; and recovery of the missing scouting skill / prior canonical ledger.

Bottom line: do not build eleven new gods. Build the perceptual-wound / silent-scar / behavior-archive machine, then let the evidence tell us whether anything deserves promotion.